

- Two players are gambling with an unfair coin (probability of head 0.3). Initially, Player 1 has \$3 and Player 2 has \$2. After each toss, Player 1 wins \$1 if head occurs and loses \$1 if tail occurs. The game ends when one player has all the money. Let X_n be the amount of money Player 1 has at "time" n . Calculate $P(X_3 = 2)$.

- Use the following data to estimate the transition probabilities and construct the Transition Probability Matrix.

1 10101 01011 10111 01011 11100 11110 10011 10110 10101

- Let $S = \{1, 2, 3\}$ and $P = \begin{pmatrix} 0.2 & 0.1 & 0.7 \\ 0.3 & 0.3 & 0.4 \\ 0.2 & 0.3 & 0.5 \end{pmatrix}$. Calculate $P(X_2 = 2 \mid X_0 = 3)$.
- There are two brands, I and II, in the market. Out of those who buy brand I in one month, 30% switch to brand II during the next month. Out of those who buy brand II in one month, 20% switch to brand I during the next month. Construct the transition probability matrix, determine the equilibrium distribution and comment.

- There are three brands in the market: 1, 2 and 3. Customer preferences are observed at the beginning of each month. The initial distribution of customer preference is $\underline{a}^{(0)} = (0.3 \ 0.3 \ 0.4)$. The TPM is:

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

Calculate the marginal distributions of X_1 and X_2 .

- Check whether the Markov Chain with the following transition probability matrix is irreducible. Classify its states and determine the periods.

$$P = \begin{pmatrix} 0.4 & 0 & 0.6 & 0 \\ 0 & 0.7 & 0 & 0.3 \\ 0 & 0.2 & 0 & 0.8 \\ 0.5 & 0 & 0.5 & 0 \end{pmatrix}$$

- Check whether the Markov Chain with the following transition probability matrix is irreducible. Classify its states and determine the periods.

$$P = \begin{pmatrix} 0 & 0.3 & 0 & 0.4 & 0.3 & 0 \\ 0 & 0.3 & 0 & 0.3 & 0.4 & 0 \\ 0.1 & 0.2 & 0.1 & 0.2 & 0.1 & 0.1 \\ 0 & 0.2 & 0 & 0.2 & 0.6 & 0 \\ 0 & 0.6 & 0 & 0.2 & 0.2 & 0 \\ 0.1 & 0 & 0.2 & 0.2 & 0.2 & 0.3 \end{pmatrix}$$